PATENT ENABLEMENT AFTER AMGEN V. SANOFI

John Moore†

TABLE OF CONTENTS

DOI: https://doi.org/10.15779/Z386H4CS2H
© 2024 John Moore
† J.D., 2025, University of California, School of Law.

I. INTRODUCTION

The U.S. patent system is built on a quid pro quo between inventors and the public. In exchange for making their inventions public and teaching persons skilled in the art how to make and use their inventions, inventors receive a limited monopoly. In *Amgen v. Sanofi*, the Supreme Court held that Amgen failed to teach antibody scientists how to make use the full scope of their claimed drug invention and therefore did not hold up its end of the bargain.[1] As a result, the Court held Amgen's patent claims invalid, thereby revoking Amgen's limited monopoly.

In 2014, the U.S. Patent and Trade Office (USPTO) granted Amgen a patent for a cholesterol-lowering antibody therapeutic. The drug works by interfering with the interaction between one of the body's chemical signals and the receptors that sense the signal. Specifically, Amgen's antibodies bind to PCSK9, the chemical signal that shuts down bad cholesterol receptors. By binding to PCSK9 at the correct location, Amgen's antibody drug blocks the interaction between PCSK9 and the bad cholesterol receptors. Without receiving the signal to stop, the bad cholesterol receptors continue to filter bad cholesterol from the blood stream, thereby lowering bad cholesterol levels.

Amgen made twenty-six of these antibodies capable of blocking PCSK9, but in its patent, Amgen claimed *all* antibodies that can block PCSK9. Specifically, the patent described Amgen's twenty-six antibodies and explained two methods by which future antibody designers could find the rest.

Sanofi made a twenty-seventh PCSK9-blocking antibody, and Amgen sued for patent infringement. The Supreme Court found that Amgen's twenty-six disclosed antibodies and two described methods of creating more did not adequately enable future drug designers to make and use *all* the antibodies Amgen claimed in its patent. The *Amgen* decision realigned patent enablement jurisprudence, which had previously used a lower enablement standard to find patent disclosures like Amgen's fully enabling, to the text of the patent statute which requires a higher enablement standard. After *Amgen*, scientific and legal limitations may force antibody patents to claim only antibodies researchers have synthesized in a lab, but upcoming technology, such as AI-powered protein modeling, may once again allow genus claims like the one used in Amgen's patent.

Part II of this Note provides the legal background necessary to understand the nuance of the Supreme Court's and lower court's decisions in *Amgen*, paying special attention to the development of the enablement doctrine from the signing of the Constitution to today. Part III delineates *Amgen's* procedural history and the Supreme Court's opinion. Part IV discusses the implications of the Court's decision and suggests that upcoming advances in technology may broaden the scope of what antibody patents can claim under *Amgen*. Part V briefly concludes.

1. Amgen Inc. v. Sanofi, 598 U.S. 594, 614 (2023).

II. THE SCOPE OF WHAT A PATENT CLAIMS MUST BALANCE WITH THE SCOPE OF WHAT A PATENT TEACHES

Amgen is a case about the bounds of the enablement doctrine. In order to discuss how the Court in *Amgen* treats the doctrine, Part II sets the scene by discussing the quid pro quo relationship central to the patent system, the different ways in which inventors claim their inventions, the development of enablement and written description from the signing of the Constitution to *Amgen*, and the specific problems that arise when patenting antibodies.

A. THE QUID PRO QUO BETWEEN INVENTORS & THE PUBLIC

Since the time of Founding Fathers, the American government has endorsed the patent as an important tool in furtherance of commerce and technological innovation. The Founding Fathers immortalized their support for patents in Article I, Section 8, Clause 8 of the Constitution: "The Congress shall have Power To . . . promote the Progress of Science and useful Arts, by securing for limited Times to Authors and Inventors the exclusive Right to their respective Writings and Discoveries."[2]

In 1790, Congress exercised its constitutional power and passed the nation's first Patent Act.[3] In this and every subsequent patent statute, Congress outlined a quid pro quo between inventors and the public.[4] In exchange for teaching skilled artisans to "make, construct, or use" an invention, the government allows the inventor a monopoly for a period of time.[5] The "period of time" for which an inventor gets their monopoly used to be fixed at seventeen years from the date of issuance. Since 1994, inventors now get twenty years of protection from the date of filing an application with the USPTO.[6]

On the other end of the bargain, under Section 112 of the 1952 Patent Act, Congress prescribed three requirements for the specification section of the patent to teach skilled artisans to "make, construct, or use" the patented invention: (1) a description of the invention, (2) how to make and use the invention, and (3) the best mode of carrying out the invention.[7] Congress

2. U.S. CONST. art. I, § 8, cl. 8.
3. 1 DONALD S. CHISUM, CHISUM ON PATENTS § 2 (2023).
4. 3 DONALD S. CHISUM, CHISUM ON PATENTS § 7.02 (2023).
5. *See Amgen*, 598 U.S. at 604–05.
6. ROBERT PATRICK MERGES & JOHN FITZGERALD DUFFY, PATENT LAW AND POLICY 68–69 (Carolina Acad. Press eds., 7th ed. (2017); 35 U.S.C. § 154 (a)(2).
7. CHISUM, *supra* note 3 § 7.01 (2023).

updated Section 112 in the 2011 America Invents Act (AIA), but left these disclosure requirements untouched.[8]

B. PATENT CLAIMS TAKE SEVERAL FORMS.

Section 112(b) of the Patent Act dictates that every inventor, after providing the public with the knowledge to make and use their invention, must then set the boundaries for what they has invented: "The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention."[9] Claims are analogous to the "metes and bounds of a real property deed."[10] Therefore, inventors aim to claim as broadly as possible to make the property they own as large as possible.[11] Inventors, however, must claim between the boundaries of what has been invented prior to their work (prior art) and what they cannot prove they invented in their specification.[12] This Note will focus on three types of claims: structural, functional, and genus.

1. *Structural Claims*

Structural claims are used when an inventor describes the elements of the invention in relation to the other claims. Consider, for example, an 1890 patent for a bicycle with an auto-adjusting v-belt pulley drive instead of the traditional chain drive.[13] In the inventor's first claim, he describes various components of his belt-drive mechanism, their interactions with each other, and their desired effects:

> Having thus described my invention, I claim—
>
> 1. The combination, with the driven pulley of a bicycle, of a self-adjusting two-part driving-pulley and a belt connecting said pulleys, whereby when resistance to forward movement is met and power is applied to overcome it the strain of the belt upon the driving pulley will cause the parts thereof to separate and the belt to move inward, thereby reducing the operative diameter of the driving pulley, so that the power is increased with a consequent loss of speed.[14]

This claim example describes all the elements one would need to create his invention with the desired effects. However, there is another way to claim an

---

8. *Id.* at § 7.02. The AIA also removed penalties for failing to satisfy the best mode requirement.
9. 35 U.S.C. § 112 (b).
10. MERGES & DUFFY, *supra* note 6 at 32.
11. *Id.*
12. *Id.*
13. U.S. Patent No. 425,390 (filed Feb. 6, 1890).
14. *Id.* at 7:26-35.

invention by listing the desired functionality without disclosing the underlying structure.

2. *Functional Claims*

Functional claiming, or the practice of describing the bounds of an invention by the result it achieves instead of its structure, was not always an accepted practice. In the 1946 case of *Halliburton v. Walker*, the Supreme Court held that Walker's patent was invalid because it described a claim element in terms of "what it will do" instead of describing its "physical characteristics" or "arrangement."[15] The Court noted, as a result of the functional nature of the claim, "there may be many other devices beyond our present information or . . . imagination" which fit the claims.[16] At its core, the Court worried that inventors were using functional claims to take more benefit (i.e., too large a scope of a claim) than what they had given with disclosure.

Congress overruled the Court, however, when it wrote functional claims into Section 112(6) of the Patent Act of 1952. Congress recognized the "difficulty with which innovation is articulated; [Section 112(6)] balances the Patent Act's requirement of particular and distinct claiming against the inherent limitations of language."[17] Section 112(6), reads:

> An element in a claim for a combination may be expressed as a means or step for performing a specified function without the recital of structure, material, or acts in support thereof, and such claim shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.[18]

There are two important takeaways from the text of the statute. First, a claim element need not demonstrate how an invention achieves a result; it need only describe the result. Second, a patent owner has a claim over all structures that produce the described result, so long as the specification describes the structures.

Suppose the inventor of the bike from above wanted to patent his belt-driven bike but, for some reason, did not want to disclose the underlying structure of the invention.[19] He could have written his claim to say the following:

---

15. Halliburton Oil Well Cementing Co. v. Walker, 329 U.S. 1, 9 (1946).

16. *Id.* at 12.

17. David J. Kappos & Christopher P. Davis, *Functional Claiming and the Patent Balance*, 18 STAN. TECH. L. REV. 365, 366 (2015).

18. 35 U.S.C. § 112(6) (1952). N.B. The America Invents Act of 2011 renumbered this passage to § 112(f).

19. *See supra* note 13 for a description of the invention.

> Having thus described my invention, I claim—
>
> 1. A method of mechanically adjusting the effective diameter of a drive pulley in combination with a static driven pulley whereby "resistance to forward movement is met and power is applied to overcome it," the mechanism adjusts, so that the power is increased with a consequent loss of speed.

In the rewritten claim above, the inventor describes all the same functionality as the structurally drafted claim but gives the reader no understanding of how he has achieved the functionality. According to the text of Section 112, so long as the inventor adequately describes the invention in the patent's description, he owns the "corresponding structure, material, or acts . . . and equivalents thereof."[20] Therefore, if the bicycle inventor adequately described the invention in the patent's specification, the claim satisfies the statutory requirement.

The mechanical fastener is another, more common example of functional language. A fastener may refer to any method of fastening two components such as a screw, bolt, nail, Velcro, glue, or welding. Mechanical patent drafters often want to cover all methods of fastening regardless of which methods may prove more practical or effective, and therefore will use functional language. Writing patents with the broad, functional language of "fastener" instead of something specific and structural like "bolt" allows mechanical patents to cover a broader scope and prevent design-arounds.

3. Genus Claims

Genus claims, which can be either structural or functional in nature, claim "individual embodiments of the invention, or species, that share a common attribute or property."[21] Similar to functional claims, genus claims offer two benefits: (1) broad scope with limited experimentation and (2) protection from design-arounds.[22] Returning to this Note's bike patent example, a genus claim might look like:

> Having thus described my invention, I claim—
>
> 1. All the species of bicycle which use a belt drive system to change drive ratios and propel the bicycle.

20. 35 U.S.C. § 112 (f).

21. Dmitry Karshtedt, Mark A. Lemley & Sean B. Seymore, *The Death of the Genus Claim*, 35 HARV. J.L. & TECH. 1, 13 (2021).

22. *Id.*

The above claim gives the inventor a monopoly to manufacture every embodiment of a bike that uses a belt drive, even if he has only created some embodiments.

But how can an inventor fulfill the quid pro quo of the patent system and provide the public with knowledge of a genus if they have only reduced to practice a handful of species? Answering this question requires a greater understanding of two additional patent requirements: enablement and written description.

C. THE PATENT STATUTE DEFINES THE SCOPE OF WHAT A PATENT'S DISCLOSURE MUST TEACH

Regardless of how one claims their invention, they must enable persons having skill in the art to practice the invention. This patent features, called enablement, gets its statutory authority from § 112(a) of the Patent Act:

> The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to **enable** any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same.[23]

To demonstrate the nuances of the statute in depth, this Note examines several cases regarding enablement. Because *Amgen* involved claims regarding both lack of enablement and written description at the district court level, Section II.C.2 also briefly highlights the differences between the doctrines of enablement and written description, as the Federal Circuit decided in *Ariad Pharmaceuticals, Inc. v. Eli Lilly and Co.*[24]

1. *Enablement Case Law*

Three cases trace the development of enablement from the drafting of the Constitution to *Amgen* at the Supreme Court in 2023. *O'Reilly v. Morse* (1853), *Consolidated Electric Light Company v. McKeesport Light Company* ("*Incandescent Lamp*") (1895), and *In re Wands* (1988) start with the statutory language in Section 112 and refine it to arrive at a modern interpretation of the enablement stature.[25] The Court in *Morse* held that, for an inventor to claim an entire technology, the inventor must enable all manifestations of the technology.[26] The Court in *Incandescent Lamp* refined enablement with respect to genus

23. 35 U.S.C. § 112 (a) (emphasis added).

24. Ariad Pharms., Inc. v. Eli Lilly & Co., 598 F.3d 1336 (Fed. Cir. 2010).

25. *See* O'Reilly v. Morse, 56 U.S. 62 (1853); Incandescent Lamp, 159 U.S. 465 (1895); *In re Wands*, 858 F.2d 731 (Fed. Cir. 1988).

26. *See O'Reilly*, 56 U.S. at 113.

claims, finding that an inventor must find "a quality common" to all members of a genus to claim the genus.[27] Finally, the Federal Circuit in *In re Wands*, found that undue experimentation precluded a properly enabled patent claim.[28] This Section will discuss each case in detail.

a) *O'Reilly v. Morse* (1853)

In *Morse*, the Supreme Court held that Morse could not claim all uses of "galvanic current" (electrical current) because he had not adequately described all uses of galvanic current.[29] Samuel Morse invented the telegraph and the eponymous Morse code in October 1832.[30] For his work, the USPTO granted Morse a patent in 1848.[31]

The same year, Morse sued O'Reilly for infringing upon his patent rights.[32] The district court found that the defendant's device's "structure and mode of operation, violated the principle of Morse's patent, and the process and means described by Morse in his specifications."[33] O'Reilly appealed, and the Supreme Court took up the case.

Upon examining Morse's patent, the Court found the first seven of eight claims valid with respect to written description and enablement.[34] But the Court found issues with Morse's eighth claim which claimed all uses of electricity:

> Eighth. I do not propose to limit myself to the specific machinery or parts of machinery described in the foregoing specification and claims; the essence of my invention being the use of the motive power of the electric or galvanic current, which I call electro-magnetism, however developed for marking or printing intelligible characters, signs, or letters, at any distances, being a new application of that power of which I claim to be the first inventor or discoverer.[35]

The Court recognized Morse's eighth claim as covering subject matter far broader than what Morse had described and enabled in his specification.[36] The

27. *See Incandescent Lamp*, 159 U.S. at 472.
28. *See In re Wands*, 858 F.2d at 736–37.
29. *Morse*, 56 U.S. at 113.
30. *Id.* at 68–69.
31. *Id.* at 112.
32. Morse v. O'Reilly, 17 F. Cas. 871, 872 (C.C.D. Ky. 1848).
33. *Id.*
34. *Morse*, 56 U.S. at 112 (finding "no well-founded objection to the description which is given of the whole invention and its separate parts, nor to his right to a patent for the first seven inventions set forth in the specification of his claims").
35. *Id.*
36. *Id.*

court interpreted Morse's patent to claim "the exclusive right to every improvement where the motive power is the electric or galvanic current, and the result is the marking or printing intelligible characters, signs, or letters at a distance."[37] Not only would this claim include any improvements on the telegraph, but it would have also later included SMS texting, television, and most of the internet. (Patent terms would have precluded Morse from claiming many of these technologies, but that is outside the scope of the argument.) Although the Court did not predict texting, television, or the internet specifically, it did find that Morse's eighth claim would let him "avail himself of new discoveries in the properties and powers of electro-magnetism which scientific men might bring to light."[38] As a sign of the times, the Court analogized Morse's claim of all "powers of electromagnetism" to that of Fulton, the inventor of "propelling vessels by steam," claiming steam-powered machines to "grind corn or spin cotton."[39]

In the end, the Court found that Morse's eighth claim was not adequately described because "he claim[ed] the exclusive right to any other mode and any other process, although not described by him, by which the end can be accomplished, if electro-magnetism is used as the motive power."[40] Thus the Court invalidated the claim.

b) *Incandescent Lamp* (1895)

In *Incandescent Lamp*, the Court held that patentees could not claim all electric lightbulbs using fibrous materials unless patentees had "discovered in fibrous and textile substances a quality common to them all, or to them generally, as distinguishing them from other materials, such as minerals, etc., and such quality or characteristic adapted them peculiarly to incandescent conductors."[41]

Sawyer and Man applied for their incandescent lamp patent in January 1880.[42] Their patent included the four following claims covering a conductor made of any fibrous or textile material, an electric circuit, a conductor made of carbonized paper, and a hermitically sealed chamber:

> (1) An incandescing conductor for an electric lamp, of carbonized fibrous or textile material, and of an arch or horseshoe shape, substantially as hereinbefore set forth.

37. *Id.*
38. *Id.* at 113.
39. *Id.*
40. *Id.* at 120–24.
41. *Incandescent Lamp*, 159 U.S. at 472.
42. *Id.* at 470.

> (2) The combination, substantially as hereinbefore set forth, of an electric circuit and an incandescing conductor of carbonized fibrous material, included in and forming part of said circuit, and a transparent, hermetically sealed chamber, in which the conductor is inclosed (sic).
>
> (3) The incandescing conductor for an electric lamp, formed of carbonized paper, substantially as described.
>
> (4) An incandescing electric lamp consists of the following elements in combination: First, an illuminating chamber made wholly of glass hermetically sealed, and out of which all carbon-consuming gas has been exhausted or driven; second, an electric-circuit conductor passing through the glass wall of said chamber, and hermetically sealed therein, as described; third, an illuminating conductor in said circuit, and forming part thereof within said chamber, consisting of carbon made from a fibrous or textile material, having the form of an arch or loop, substantially as described, for the purpose specified.[43]

These claims ultimately covered a conductor made of *any* fibrous or textile material, an electric circuit, a conductor made of carbonized paper, and a hermitically sealed chamber. But their specification included little guidance as to what *kind* of fibrous or textile materials one should use to create an incandescent light.[44] Sawyer and Man described using carbonized paper and wood carbon as conductors and discussed the advantages of using "vegetable fibrous or textile material instead of mineral or gas carbon."[45] Yet, there are innumerable types of vegetable fibrous or textile materials, and Sayer and Man provided no guidelines for future incandescent lamp makers to pick one from the universe of possible materials.[46]

Sawyer and Man asserted their patent rights against the McKeesport Light Company, which was using incandescent lamps described in Thomas Edison's patents.[47]

The Court, comparing Edison's description of his lightbulb to Sawyer and Man's, held that Sawyer and Man's patent was "so vague and uncertain that no one can tell, except by independent experiments, how to construct the

43. *Id.* at 468.

44. *See id.* at 466–68.

45. *Id.* at 467–68.

46. *Id.* at 472 (noting, for example, that "an examination of over 6,000 vegetable growths showed that none of them possessed the peculiar qualities that fitted them for [incandescent conductors].").

47. *Id.* at 465.

patented device."[48] Because Sawyer and Man had found and described only one fibrous or textile material which suited their purpose, they did not have the right to "exclude everybody from the whole domain of fibrous and textile materials."[49] The Court extended its logic to say that "the fact that paper happens to belong to the fibrous kingdom did not invest them with sovereignty over this entire kingdom, and thereby practically limit other experimenters to the domain of minerals."[50] Finally, Sawyer and Man's patent would exclude experimentation that would improve on what Sawyer and Man had invented.[51] Therefore, the Court held all but the third claim specifically regarding carbonized paper were too broad and invalid.[52]

c) *In re Wands* (1988)

The Federal Circuit in *In re Wands*, outlined eight factors to determine if a patent's disclosure would adequately enable a person having skill in the art (PHOSITA) to practice an invention without undue experimentation:

> (1) the quantity of experimentation necessary, (2) the amount of direction or guidance presented, (3) the presence or absence of working examples, (4) the nature of the invention, (5) the state of the prior art, (6) the relative skill of those in the art, (7) the predictability or unpredictability of the art, and (8) the breadth of the claims.[53]

In this case, Jack Wands appealed the USPTO's § 112 rejection of his application for a patent describing an antibody that detects Hepatitis B.[54] The claim at issue reads:

> 1. An immunoassay method utilizing an antibody to assay for a substance comprising hepatitis B-surface antigen (HBsAg) determinants which comprises the steps of:
>
> contacting a test sample containing said substance comprising HBsAg determinants with said antibody; and
>
> determining the presence of said substance in said sample;

48. *Id.* at 474.
49. *Id.* at 476.
50. *Id.*
51. *Id.*
52. *Id.* at 477.
53. *In re Wands*, 858 F.2d at 737.
54. *Id.* at 733.

> wherein said antibody is a monoclonal high affinity IgM antibody having a binding affinity constant for said HBsAg determinants of at least 109 M^{-1}.[55]

The USPTO had rejected the above claim on the grounds that the patent's specification did not enable a person having skill in the art to practice the invention.[56] Wands argued that the claims were fully enabled "because the monoclonal antibodies needed to perform the immunoassays can be made from readily available starting materials using methods that are well known in the monoclonal antibody art."[57]

Here the Federal Circuit found that "undue experimentation would not be required to practice the invention."[58] The court started its reasoning by reiterating that enablement was not precluded by the necessity for experimentation: only *undue* experimentation, determined by "weighing many factual considerations."[59] The court relied on the facts that (1) Wands used commercially available radioimmunoassay kits to screen for cells that produced antibodies directed against Hepatitis B, (2) the screening techniques Wands used were "well known in the monoclonal antibody art," and (3) four of the nine antibody producing cells, "hybridomas," produced antibodies that bound to Hepatitis B within the patent's parameters."[60] The *Wands* factors are often cited at the district court level when determining whether a patent's specification would require undue experimentation to enable a PHOSITA to practice the full scope of the claims.

2. *The Federal Circuit Clarified that Enablement & Written Description are Separate Requirements under § 112(a)*

As noted above, § 112(a) requires that a patent's specification include a "written description of the invention and of the manner and process of making and using it . . . as to enable any person skilled in the art . . . to make and use the same."[61] Litigants and courts have expressed confusion whether the statute has two separate requirements—written description and enablement—or one overarching requirement. The Federal Circuit tackled this statutory interpretation question in *Ariad Pharmaceuticals, Inc. v. Eli Lilly & Co.*[62]

55. *Id.* at 734.
56. *Id.* at 734–35.
57. *Id.* at 736.
58. *Id.* at 736–37.
59. *Id.*
60. *Id.* at 738–40.
61. 35 U.S.C. § 112(a).
62. *Ariad*, 598 F.3d at 1343.

On one side of the argument, Ariad argued that the statute required two components: a written description of the invention and a written description of the manner and process of using it.[63] Under this interpretation, the prepositional phrase of § 112 "in such full, clear, concise, and exact terms as to enable any person skilled in the art . . . to make and use the same," applied to both written descriptions.[64] On the other side of the argument, Eli Lilly argued the statute required three components: a written description of the invention; a written description of the manner and process of using it in such full, clear, concise, and exact terms as to enable any person skilled in the art to make and use the same; and the best mode contemplated by the inventor of carrying out the invention.[65]

The Federal Circuit ruled that there are "two separate description requirements."[66] But the court found that the enablement "prepositional phrase, in such full, clear, concise, and exact terms as to enable any person skilled in the art . . . to make and use the same" modifies only the written description of the manner and process of making and using the invention.[67] Therefore, the "adequacy of the description of the manner and process of *making* and *using* the invention is judged by whether that description enables one skilled in the art to *make* and *use* the same."[68] This then begs the question of what defines an adequate written description of the invention. The court found that the specification "must describe an invention understandable to that skilled artisan and show that the inventor actually invented the invention claimed."[69]

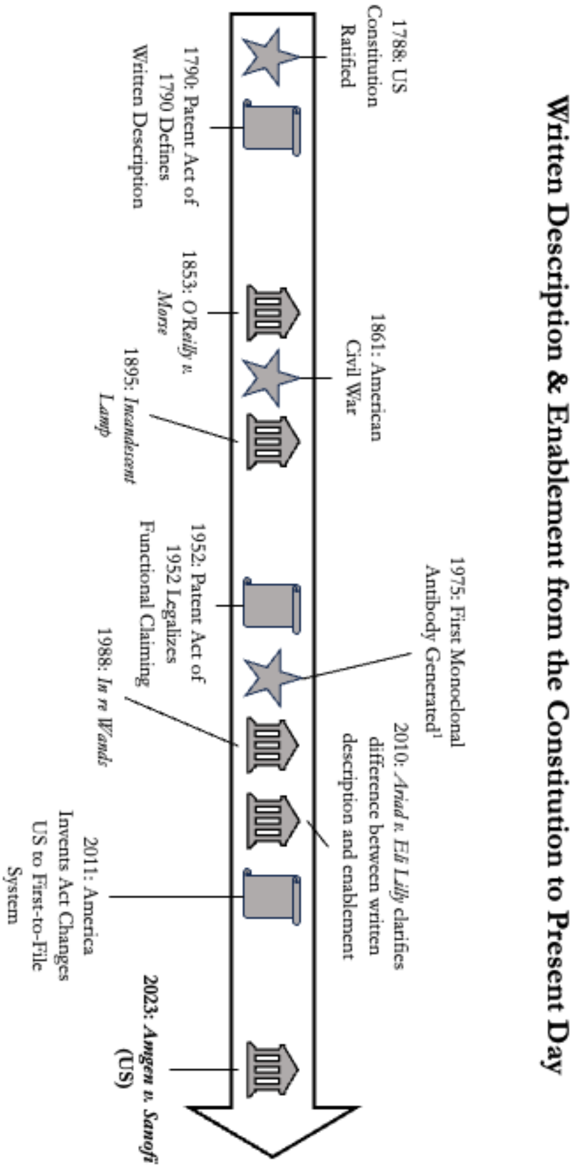
In sum, the Federal Circuit ruled that a specification contains two requirements: (1) a written description of the invention that is "understandable to that skilled artisan and show that the inventor actually invented the invention claimed," and (2) a written description of how to make and use the invention "in such full, clear, concise, and exact terms as to enable any person skilled in the art . . . to make and use the same."[70] The first requirement is referred to as written description in practice, and the second requirement is referred to as enablement. Because the Federal Circuit only elaborated on the difference between written description and enablement in *Ariad* in 2010,

63. *Id.*
64. *Id.*
65. *Id.* at 1344.
66. *Id.*
67. *Id.* (internal quotations omitted).
68. *Id.*
69. *Id.* at 1351.
70. *See generally id.*

decisions before *Ariad* generally treat written description and enablement as one requirement and discuss them concurrently.

3. *Timeline*

Below is a non-exhaustive timeline to help visualize the progress of written description and enablement through the jurisprudence of patent law.



D. ANTIBODY PATENTS DO NOT EASILY FIT INTO THE ENABLEMENT FRAMEWORK

The patent at issue in *Amgen* is for a cholesterol-lowering antibody therapeutic. However, since the first synthesized antibody in 1975 and until *Amgen* in 2023, antibody patents have operated under different enablement standards because antibodies do not fit nicely into normal enablement doctrine. This different enablement standard is known as the antibody exception. This Section first shows the scientific and legal backgrounds for the antibody exception, next defines the bounds of the exception at its strongest, and finally outlines its downfall.

1. *Scientists View Antibodies Functionally*

Viewing antibodies from a scientist's perspective will illuminate why scientists define antibodies by the function they perform instead of the molecules that make up the antibody. This will in turn help illustrate the historic difficulty with writing antibody claims.

Antibodies are relatively large molecules naturally found in the body as part of the immune system.[71] When the immune system encounters an antigen—"[a]ny molecule or pathogen capable of eliciting an immune response" such as a "virus, a bacterial cell wall, or an individual protein or other macromolecule,"—the immune system produces antibodies to bind to the surface of the antigen in key places so that the antigen cannot interact with the body and cause harmful immune consequences.[72]

When the body responds to an antigen, it produces "several different antibodies" that bind to the antigen.[73] Each type of antibody may bind to a different site on the antigen; there are even different antibodies that can bind to the same site on an antigen.[74] In this way, antigens are like locks with several keyholes, each of which can accept many different keys. Because of the wide variety of antibodies that can successfully bind to an antigen, scientists think about antibodies functionally. It is more common to discuss the antibodies that develop as part of the immune system response to the influenza virus, for example, than it is to discuss a singular antibody that binds to the virus.

But that is not to say that scientists cannot discuss an individual antibody. Scientists describe the structure of an individual antibody, like all proteins, via its primary structure, which refers to the two-dimensional chain of amino acids

71. DAVID L. NELSON & MICHAEL M. COX, PRINCIPLES OF BIOCHEMISTRY 165 (Lisa Kinne et al. eds., 8th ed. 2021).

72. *Id.*

73. *Id.*

74. *Id.*

that make up the antibody, and its tertiary structure, which refers to the three-dimensional shape of the antibody.[75] The three-dimensional shape of the binding region of the antibody determines to what antigen or antigens a particular antibody will bind.[76]

This scientific manner of thinking about antibodies as a genus with a particular function naturally extends into approaches for patenting antibodies. Such scientists who view their inventions functionally want coverage for the full genus of antibodies that can cause a particular effect (e.g., blocking an antigen).

2. *The Functional Nature of Antibodies Leads to Trouble with Patenting*

An antibody patent applicant must choose between a patent with broad scope that prevents design-arounds and a patent that is likely to be granted. For example, if an inventor lists in the patent specification the chains of amino acids that make up the antibody, a PHOSITA should be able to synthesize the antibody with the same sequence, and thus the invention is perfectly enabled/described.[77] And the patent office should have no problem granting the patent. However, using the methods of conservative substitution,[78] another scientist can easily design around the first scientist's patent. While the patent application will likely be granted, the inventor does not own a strong patent because they cannot stop others from easily designing around their invention.

Alternatively, if a scientist attempts to patent the entire genus of antibodies that bind to a particular antigen (which could encompass hundreds, thousands, or perhaps millions of antibodies), the USPTO will be hesitant to grant such a patent without "a quality common to" the entire genus.[79] Because antibodies with widely varying amino acid sequences can function similarly, finding the

75. *Id.* at 90.

76. *Id.* at 167.

77. While it is possible to describe an antibody by its structure, it is largely impractical. Some legal scholars have analogized describing antibodies by their atomic structure to "describing a fighter jet by listing every nut and bolt." Therefore, patent owners often try to claim antibodies functionally, often describing a genus of antibodies which fulfill a particular function. *See* Mark Lemley & Jacob Sherkow, *The Antibody Patent Paradox*, 132 YALE L.J. 994, 998 (2023).

78. Conservative substitution is "amino acid replacement in a protein that changes a given amino acid to a different amino acid with similar biochemical properties." Mingrui Wang, Dapeng Wang, Jun Yu & Shi Huang, *Enrichment in Conservative Amino Acid Changes Among Fixed and Standing Missense Variations in Slowly Evolving Proteins*, PeerJ 2 (Sept. 16, 2020).

79. *See O'Reilly*, 56 U.S. at 120–24.

thread that connects the antibodies in the genus is impossible except for describing their functionality.[80]

Therefore, scientists are left with a challenging dilemma: should the patents claim specific sequences of amino acids, or describe antibodies purely by describing the antigen they bind to? The Federal Circuit provides some guidance.

3. *The Federal Circuit Permitted Lower Written Description and Enablement Standards for the Nascent Field of Antibody Technology*

The Federal Circuit has often allowed "product-by-process" claims.[81] These claims meet the enablement requirement by allowing inventors to describe (1) a process through which PHOSITAs put certain starting materials, and (2) the resulting product.[82] These claims are particularly useful when an inventor cannot "describe the structure of the resulting product."[83] Because the first antibody scientists did not have a complete understanding of antibodies, early antibody patents operated under this scheme.[84]

For example, in *Hybritech Incorporated v. Monoclonal Antibodies*, the Federal Circuit found that Hybritech's product-by-process antibody claim satisfied the enablement requirement.[85] Hybritech simply claimed their process by which a PHOSITA might use an antibody to determine the presence of an antigen. The claim at issue read:

> 1. A process for the determination of the presence of [sic, or] concentration of an antigenic substance in a fluid comprising the steps:[86]

The exact steps described in the claim are not important, but the court's finding of enablement shows that the Federal Circuit was not uncomfortable with product-by-process claims.

But functional claiming, including product-by-process claiming, requires increased disclosure under the patent statute.[87] In *The Antibody Patent Paradox*, Mark Lemley and Jacob Sherkow describe problems with functionally claiming

80. *See Incandescent Lamp*, 159 U.S. at 472 (finding a genus claim regarding fibrous and textile materials for incandescent conductors which did not identify a quality "common to them all" too broad).
81. Lemley & Sherkow, *supra* note 77 at 1016–17.
82. *Id.*
83. *Id.* at 1017.
84. *Id.*
85. Hybritech Inc. v. Monoclonal Antibodies, Inc., 802 F.2d 1367, 1384 (Fed. Cir. 1986).
86. *Id.* at 1370.
87. 35 U.S.C. § 112(f).

antibodies.[88] For example, antibody genus claims can be too broad: claims "specific to nothing more than a single antigen would encompass *every* antibody that happened to bind to it—potentially thousands, if not millions, of molecules."[89] These claims would be invalid for multiple reasons, including anticipation by prior art of a single pre-existing antibody that binds to the specified antigen.[90] Instead, patentees could describe how the genus of antibodies bind to a particular epitope, or region of an antigen—this has historically been a specific enough disclosure for the PTO to grant a patent.[91] Patentees could also narrow claims by describing the affinity, or binding strength, between the genus of antibodies and the antigen.[92]

For example, in *In re Wands*, the patent only described the binding affinity between the antibody genus and the Hepatitis B surface antigen.[93] Wands claimed:

> 1. An immunoassay method utilizing an antibody to assay for a substance comprising hepatitis B-surface antigen (HBsAg) determinants which comprises the steps of:
>
> contacting a test sample containing said substance comprising HBsAg determinants with said antibody; and
>
> determining the presence of said substance in said sample;
>
> wherein said antibody is a monoclonal high affinity IgM antibody having a binding affinity constant for said HBsAg determinants of at least 109 M^{-1}.[94]

The Federal Circuit determined that "undue experimentation would not be required to practice the invention," and therefore Wands's functional claim was enabled.[95] Because the Federal Circuit supported functional claims for antibodies in various rulings, the USPTO responded by releasing new guidance to its examiners.

4. *The Antibody Exception a.k.a. the Newly Characterized Antigen Test*

The antibody exception, also known as the newly characterized antigen test, reached its apex when, in 1999, the USPTO published the Revised Written

88. Lemley & Sherkow, *supra* note 77 at 1013.
89. *Id.*
90. *Id.*
91. *Id.* at 1014.
92. *Id.*
93. *In re Wands*, 858 F.2d at 734–40.
94. *Id.* at 734.
95. *Id.* at 738–40.

Description Guidelines Training Materials ("1999 Training Materials").[96] In the Training Materials, the USPTO instructed patent examiners that antibody function claims supported by relatively sparse written description meet the written description requirement under § 112.[97]

As a part of the guidelines, the USPTO included several example patent specifications. In one example pertaining to a hypothetical antibody patent, the USPTO outlines a specification teaching that "antigen X has been isolated and is useful for detection of HIV infections."[98] The example specification only adds the antigen's molecular weight and the protocol to isolate the antigen.[99] The example specification also "contemplates but does not teach in an example antibodies which specifically bind to antigen X."[100]

The example claim then reads in its entirety, "[a]n isolated antibody capable of binding to antigen X."[101] The USPTO's guidelines on written description indicate that this antibody function claim, supported only by vague contemplations in the patent specification, "meets the requirement under 35 USC 112 first paragraph as providing an adequate written description of the claimed invention."[102]

In 2008, the USPTO published re-revised guidelines ("2008 Training Materials") to its patent examiners that retained the antibody exception.[103] The 2008 Training Materials similarly outline a case in which the specification does not describe any of the following: "the complete structure of an antibody capable of binding antigen X," "a partial structure of the claimed antibody," "any physical or chemical properties of the claimed antibody," "a correlation between the function of binding to antigen X and the structure of the claimed antibody," and "a method of making an antibody that binds antigen X."[104] The

96. *See* Comm'r Pat., Revised Written Description Guidelines Training Materials (1999), https://web.archive.org/web/20110928022825/http://www.uspto.gov/web/offices/pac/writtendescription.pdf.

97. *See id.* at 59–60.

98. *Id.* at 59.

99. *Id.*

100. *Id.*

101. *Id.*

102. *Id.* at 60. N.B. The 1999 Training Materials nor the 2008 Training Materials will differentiate between the enablement and written description requirements because the USPTO published the materials before *Ariad*, 598 F.3d 1336, 1351 (Fed. Cir. 2010) (finding § 112 requires both a written description of the invention which is "understandable to that skilled artisan and show that the inventor actually invented the invention claimed" and a written description of how to make and use the invention "in such full, clear, concise, and exact terms as to enable any person skilled in the art . . . to make and use the same").

103. Comm'r Pat., Written Description Training Materials, Revision 1 (2008), https://www.uspto.gov/sites/default/files/web/menu/written.pdf.

104. *Id.* at 45–46.

specification meets the written description requirement by simply stating that "one of skill in the art would have recognized that the disclosure of the adequately described antigen X put the applicant in possession of antibodies which bind to antigen X."

The USPTO's Training Materials shortly appeared in Federal Circuit litigation. For example, in *Enzo Biochem v. Gen-Probe* (2002), the Federal Circuit cited the antibody exception as laid out in the 1999 Training Materials.[105] Having stated that the 1999 Training Materials were only persuasive and not binding, the court nevertheless applied the logic of the antibody example discussed above when deciding the written description standard for Enzo's nucleic acid probe patent.[106]

5. Moving Away from the Antibody Exception

According to the Federal Circuit, the first time that court "examined the newly characterized antigen test in some detail" was *Centocor* in 2011.[107] In *Centocor*, the Federal Circuit clarified the antibody exception: although "precedent suggests that written description for certain antibody claims can be satisfied by disclosing a well-characterized antigen, that reasoning applies to disclosure of newly characterized antigens where creation of the claimed antibodies is routine."[108] The court reasoned that, although Centocor described characteristics such as "high affinity, neutralizing activity, and A2 specificity" of the genus of claimed antibodies, obtaining antibodies that met these requirements was impossible using "conventional, routine, well developed and mature" methods.[109] The court concluded that Centocor failed to show that "generating fully-human antibodies with the claimed properties would be straightforward for a person of ordinary skill in the art given the state of human antibody technology," and therefore, the invention was not adequately described.[110]

Here, the court's analysis muddled two standards. In language very similar to the "undue experimentation" enablement standard from *In re Wands*, the court looked at whether it would be "straightforward for a person of ordinary skill in the art" to generate the claimed antibodies.[111] But the court used this test in its written description analysis.[112] This Note is not the first to point out

105. Enzo Biochem, Inc. v. Gen-Probe Inc., 323 F.3d 956, 964 (Fed. Cir. 2002).
106. *Id.*
107. *See* Amgen v. Sanofi, 872 F.3d 1367, 1377 (Fed. Cir. 2017).
108. Centocor Ortho Biotech, Inc. v. Abbott Lab'ys, 636 F.3d 1341, 1352 (Fed. Cir. 2011).
109. *Id.*
110. *Id.* at 1352–53.
111. *Id.*; *see In re Wands*, 858 F.2d at 736–37.
112. *Centocor*, 636 F.3d at 1352.

this anomaly: Lemley and Sherkow also noted that "the Federal Circuit's antibody jurisprudence [had] started to conflate enablement with written description."[113]

Lemley and Sherkow believe the first time the Federal Circuit took steps to dismantle the antibody exception was in *Chiron Corp. v. Genentech, Inc.*, in 2004.[114] In *Chiron*, Chiron claimed a breast cancer antibody-treatment: "A monoclonal antibody that binds to human c-erbB-2 antigen."[115] The court found that the claim included both "murine but also chimeric antibodies," but although "Chiron's applications certainly enable murine antibodies, they do not enable chimeric antibodies."[116] The court further found that "Chiron's disclosure fell short of providing a specific and useful teaching of all antibodies within the scope of the claim."[117] Here, the Federal Circuit did not outright reject the newly characterized antigen test. The presence of bad facts in the case, namely trying to argue that murine antibody examples would enable a PHOSITA to make chimeric antibodies, discourages the conclusion that the Federal Circuit was trying to turn away from functional claiming instead of punishing sloppy claiming.[118]

In addition to *Centocor*, the Federal Circuit in *AbbVie* also made "a fundamental shift away from the functional characterization of antibodies."[119] In *AbbVie*, the court found that functional genus claims were "inherently vulnerable to invalidity challenge for lack of written description support, especially . . . where it is difficult to establish a correlation between structure and function for the whole genus."[120] Specifically, the Federal Circuit held that AbbVie's claim over a genus of psoriasis and rheumatoid arthritis preventing antibodies was invalid for lack of written description because "AbbVie's patents only describe one type of structurally similar anti-bodies (sic) and that

113. Lemley & Sherkow, *supra* note 77 at 1020.

114. *Id.*

115. Chiron Corp. v. Genentech, Inc., 363 F.3d 1247, 1250 (Fed. Cir. 2004).

116. *Id.* at 1256. Murine antibodies are those produced naturally in mice. Chimeric antibodies are those genetically engineered from antibodies produced by two or more species: here, humans and mice.

117. *Id.* (internal citations omitted).

118. *See id.* at 1256. In addition to over-broad claiming in comparison to enablement, Chiron also sought to claim priority to its 1984 patent application in its 1986 patent application, which included chimeric antibodies, even though chimeric antibodies "did not even appear for the first time until several months after the 1984 application." *Id.* at 1257–58.

119. Lemley & Sherkow, *supra* note 77 at 1028.

120. AbbVie Deutschland GmbH & Co., KG v. Janssen Biotech, Inc., 759 F.3d 1285, 1301 (Fed. Cir. 2014).

those antibodies are not representative of the full variety or scope of the genus."[121] AbbVie's representative claim in issue read:

> 29. A neutralizing isolated human antibody, or antigen-binding portion thereof that binds to human IL-12 and dissociates from human IL-12 with a k_{off} rate constant of 1×10^{-2} s^{-1} or less, as determined by surface plasmon resonance.[122]

The court analogized AbbVie's genus claims to a plot of land: "if the disclosed species only abide in a corner of the genus, one has not described the genus sufficiently to show that the inventor invented, or had possession of, the genus. He only described a portion of it."[123] By only describing one corner of the metaphorical plot of land, AbbVie had outlined a "research plan" and not described the genus.[124]

The ruling in *AbbVie* was a significant departure from the earlier standard outlined in *In re Wands* and the 1999 Training Materials where merely describing the antigen and contemplating some example antibodies was adequate written description and enablement. Table 1 traces and summarizes the enablement and written description standards through the years. With *AbbVie* as a backdrop, this Note now shifts towards *Amgen*.

Table 1: Table Outlining Developments in the Rise and Fall of the Antibody Exception / Newly Characterized Antigen Test

Federal Circuit Case / USPTO Document / Event	Year	Impact on the Antibody Exception
First Monoclonal Antibody Generated[125]	1975	N/A
In re Wands	1988	Describing an antigen and the binding affinity between the antibody and antigen is enough to fully enable a PHOSITA to practice the scope of the claimed invention, a genus of antibodies.

121. *Id.* at 1298–1300.
122. *Id.* at 1292.
123. *Id.* at 1300.
124. *See id.*
125. Justin K.H. Liu, *The History of Monoclonal Antibody Development – Progress, Remaining Challenges and Future Innovations*, 3 ANNALS MED. & SURGERY 113, 1 (2014).

USPTO Written Description Training Materials	1999	Characterization of an antigen and contemplation of example antibodies is enough to describe a genus of antibodies.
Chiron v. Genentech	2004	Describing an antigen and describing only murine antibodies does not enable a PHOSITA to make and use chimeric antibodies.
Revised USPTO Written Description Training Materials	2008	Describing an antigen with its amino acid sequence and discussing antibodies which bind to the antigen is enough to describe a genus of antibodies.
Centocor v. Abbott	2011	Describing only an antigen and antibodies that bind to it with certain properties is not enough to describe a genus of antibodies such that a PHOSITA would be able to generate the described antibodies in a straightforward manner.
AbbVie v. Janssen	2014	Describing an antigen, outlining a genus of antibodies that bind to it with specific characteristics, and providing several examples is not enough to adequately describe the genus if there was not a description of enough representative species to describe the diversity of the genus.

III. AMGEN V. SANOFI

This Part of the Note traces *Amgen*'s path starting in district court, moving to the Federal Circuit, going back to the district court on remand, appealing again to the Federal Circuit, and finally landing at the Supreme Court.

A. PROCEDURAL POSTURE

1. *District of Delaware I*

On October 17, 2014, Amgen filed a complaint against Sanofi for patent infringement in the District of Delaware.[126] Amgen alleged that Sanofi was currently or imminently manufacturing or selling an anti-PCSK9 antibody, a cholesterol drug, which infringed on Amgen's patent.[127] Amgen claimed every antibody such that, "when bound to PCSK9, the monoclonal antibody binds to at least one of the following residues . . . and wherein the monoclonal antibody blocks binding of PCSK9 to LDL[-]R."[128]

Sanofi responded by claiming Amgen's patents were invalid.[129] Eighteen months later, on March 16, 2016, a jury found that Amgen's patents were, in fact, not invalid.[130] The court denied Sanofi's subsequent renewed motion for judgment as a matter of law (JMOL) that Amgen's patents were invalid due to lack of written description and enablement.[131] The court cited expert testimony that "it would not be possible to start with the amino acid sequences listed in the specification and make the full diversity of antibodies that are covered by the claims, because it's a very unpredictable process and would require trial and error."[132] The court granted Amgen's motion for a permanent injunction against Sanofi; Sanofi appealed to the Federal Circuit.[133]

2. *Federal Circuit I*

The Federal Circuit found that the trial court "improperly excluded evidence regarding written description and enablement," inter alia, reversed in part and remanded for a new trial to determine written description and enablement.[134]

The court also took the opportunity to state that the "newly characterized antigen test flouts basic legal principles of the written description

126. Compl. for Patent Infringement and Declaratory J. of Patent Infringement, 20, Amgen Inc. v. Sanofi, No. 14-01317 (D.D.E. 2016).

127. *Id.* at 1.

128. *Amgen*, 872 F.3d at 1372.

129. Answer to Compl. with Jury Demand at 10, Amgen Inc. v. Sanofi, No. 14-01317 (D.D.E. 2016) ECF No. 20.

130. Redacted Version of 303 Jury Verdict at 1–2, Amgen Inc. v. Sanofi, No. 14-01317 (D.D.E. 2016) ECF No. 304.

131. Mem. Op. at 29, Amgen Inc. v. Sanofi, No. 14-01317 (D.D.E. 2016) ECF No. 389.

132. *Id.* at 17.

133. Mem. Order at 7, Amgen Inc. v. Sanofi, No. 14-01317 (D.D.E. 2016) ECF No. 392; Notice of Appeal at 1, Amgen Inc. v. Sanofi, No. 14-01317 (D.D.E. 2016) ECF No. 402.

134. *Amgen*, 872 F.3d at 1371.

requirement."[135] The court explained that the test "contradicts the statutory quid pro quo" by allowing "patentees to claim antibodies by describing something that is not the invention, i.e., the antigen."[136]

3. District of Delaware II

Back in the district court, after another trial, Sanofi filed a motion for JMOL that the patent neither meets written description or enablement requirements. On the issue of written description, Sanofi argued that "no reasonable jury could conclude that [Amgen's] claims are supported by written description under either the representative species test or the structural features test."[137] The court denied the motion in part, finding that "substantial evidence in the record supports the jury verdict of validity under the representative species test" for written description.[138]

On the issue of enablement, Sanofi also argued that Amgen's claims were not enabled because "the vast majority of antibodies within the full scope of the claims are impossible to make" and "undue experimentation is required to make antibodies within the claimed genus."[139] Regarding the first claim, the court determined that the evidence did not show that the claimed embodiments were impossible to make.[140] As to the second claim, the court looked at the *Wands* factors to determine that "undue experimentation would be needed to practice the full scope of the invention."[141] The court noted that because the "number of antibodies potentially falling within the claim scope is in the millions," and that "substitution of amino acids in a sequence may have unpredictable effects on the function of the antibody."[142] The court concluded that "despite the routine techniques employed [to identify antibodies within the scope of the genus], it appears that a person of ordinary skill in the art would still be required to do essentially the same amount of work as the inventors of the patents-in-suit."[143] Therefore, the court found that practicing the full scope of the claims would require undue experimentation, granted Sanofi's motion for JMOL for lack of enablement, and found Amgen's patent claims invalid.[144]

---

135. *Id.* at 1378.
136. *Id.*
137. Mem. Op. at 6, Amgen Inc. v. Sanofi, No 14-01317 (D.D.E. 2019) ECF No. 1050.
138. *Id.* at 9.
139. *Id.* at 10.
140. *Id.* at 12–13.
141. *Id.* at 24.
142. *Id.* at 23.
143. *Id.*
144. *Id.* at 24.

Amgen appealed the results of the JMOL for lack of enablement.[145]

4. *Federal Circuit II*

On February 11, 2021, the Federal Circuit affirmed the lower court's decision on enablement.[146] The court revisited the *Wands* factors to determine whether Amgen's patent was invalid due to undue experimentation.[147] The court also analogized to several cases in which similarly, "the specification failed to teach one of skill in the art whether the many embodiments of the broad claims would exhibit that required functionality."[148] The court continued by noting that each claim on appeal was a functional claim.[149] Agreeing with the district court, the Federal Circuit found that "the specification here did not enable preparation of the full scope of these double-function claims without undue experimentation."[150] The Federal Circuit, returning to the *Wands* factors, concluded that, "[t]he functional limitations here are broad, the disclosed examples and guidance are narrow, and no reasonable jury could conclude under these facts that anything but 'substantial time and effort' would be required to reach the full scope of claimed embodiments."[151]

Amgen petitioned for a writ of certiorari which was granted on November 4, 2022.[152]

B. THE SUPREME COURT FOUND AMGEN'S PATENT CLAIMS INVALID FOR LACK OF ENABLEMENT

> In exchange for bringing new designs and technologies into the public domain through disclosure, so they may benefit all, an inventor receives a limited term of protection from competitive exploitation.[153]

The Court started its opinion by reiterating the quid pro quo central to the institution of the US patent system: in exchange for disclosing their invention to the public enabling a person skilled in the art to make and use an invention,

145. Notice of Appeal, Amgen Inc. v. Sanofi, No 14-01317 (D.D.E. 2019) ECF No. 1063.
146. Amgen Inc. v. Sanofi, Aventisub LLC, 987 F.3d 1080, 1082 (Fed. Cir. 2021).
147. *Id.* at 1086.
148. *Id.*; *see* Wyeth & Cordis Corp. v. Abbott Laboratories, 720 F.3d 1380, 1385–86 (Fed. Cir. 2013); *see also* Enzo Life Sciences, Inc. v. Roche Molecular Systems, Inc., 928 F.3d 1340, 1345–48 (Fed. Cir. 2019); Idenix Pharmaceuticals LLC v. Gilead Sciences Inc., 941 F.3d 1149, 1160–63, 1165 (Fed. Cir. 2019).
149. *Amgen*, 987 F.3d at 1087.
150. *Id.*
151. *Id.* at 1088.
152. Pet. for Writ of Cert., Amgen Inc. v. Sanofi, Aventisub LLC, No. 20-1074 (Fed. Cir. 2021).
153. *Amgen*, 598 U.S. at 604 (internal quotations omitted).

the patent owner receives a limited monopoly.[154] And so the Court took up the question of whether Amgen had held up its end of the bargain.[155]

The Court started its analysis using the first principles of patent law. It described the statutory support for written description and enablement.[156] It then listed and explained key enablement cases—*Morse*, *Incandescent Lamp*, and *Holland Furniture*—in which the patent at issue did not enable the full scope of its claims.[157] The Court contrasted those cases with ones such as *Wood* and *Minerals Separation*, in which the patent's specification was not "necessarily inadequate just because it [left] the skilled artist to engage in some measure of adaptation or testing."[158] Differentiating between the two sets of cases, the Court found that "[d]ecisions such as Wood and Minerals Separation establish that a specification may call for a reasonable amount of experimentation to make and use a patented invention. What is reasonable in any case will depend on the nature of the invention and the underlying art."[159] The Court condensed the jurisprudence here: "For if our cases teach anything, it is that the more a party claims, the broader the monopoly it demands, the more it must enable."[160]

The Court analogized Amgen's antibody patent enablement to an enablement for a lock with one hundred tumblers and twenty positions each.[161] If the enablement section for a patent on a method of unlocking the lock listed twenty-six successful combinations and gave instructions on how to find more amounting to "randomly try[ing] a large set of combinations and then record[ing] the successful ones," the Court found that "all successful lock combinations" would not have been enabled.[162] Similarly, the Court found that Amgen's antibody claim was not fully enabled because it called "on scientists to create a wide range of candidate antibodies and then screen each to see which happen to bind to PCSK9 in the right place and block it from binding to LDL receptors."[163] Although Amgen disclosed two methods, the "roadmap" and "conservative substitution" methods, to make the rest of the

154. *See id.* at 604–05.
155. *Id.* at 599.
156. *Id.* at 604–05; *see also supra* notes 2, 20.
157. *Amgen*, 598 U.S. at 606–10; *see also supra* Section II.D.2.
158. *Amgen*, 598 U.S. at 611–12.
159. *Id.* at 612.
160. *Id.* at 613.
161. *Id.* at 614.
162. *Id.*
163. *Id.*

genus of antibodies, the Court found these amounted to "little more than two research assignments."[164]

Finding that the Federal Circuit correctly identified that Amgen's patent claims were not fully enabled, the Supreme Court affirmed.[165]

IV. *AMGEN* DISCOURAGED TECH-EXCEPTIONALISM AND REALIGNED ENABLEMENT DOCTRINE WITH THE PATENT STATUTES, BUT NEW TECHNOLOGY COULD STILL ENABLE BROAD ANTIBODY GENUS CLAIMS

Part IV will examine the legal analysis the Supreme Court and Federal Circuit used to find Amgen's patent claims not enabled, discuss how *Amgen* will affect enablement moving forward, and explore how developing technology may allow broad antibody genus claims to be enabled after *Amgen*.

A. *AMGEN* DISCOURAGES TECH-EXCEPTIONALISM AND FAVORS A TEXTUALIST READING OF THE ENABLEMENT STATUTE

The Supreme Court in *Amgen* broadly discouraged tech-exceptionalism, including the antibody exception, and endorsed the full-scope enablement test as the enablement standard for all technologies moving forward. Section IV.A.1 will argue that, under the Supreme Court's textualist interpretation of the enablement statute, the antibody exception is no longer consistent with case law. Then, Section IV.A.2 will show how the Court discouraged tech-exceptionalism in its *Amgen* opinion. Finally, Section IV.A.3 will examine the full scope enablement test and demonstrate how the Federal Circuit has previously applied it.

1. *The Antibody Exception is Inconsistent with the Court's Amgen Opinion*

As Part II discussed, the USPTO's 1999 training materials allowed patent examiners to evaluate patents using a standard of enablement lower than the standard prescribed in the text of § 112 when examining an antibody genus claim.[166] Amgen's 2014 patent surpassed the 1999 training materials standard.

164. *Id.*

165. *Id.* at 616.

166. *See* Comm'r Pat., Revised Written Description Guidelines Training Materials 59 (1999), https://web.archive.org/web/20110928022825/http://www.uspto.gov/web/offices/pac/writtendescription.pdf. The training materials described an example patent whose specification "contemplate[d] but [did] not teach in an example antibodies which specifically bind to antigen X" and whose sole claim read "[a]n isolated antibody capable of binding to antigen X." Under these guidelines, the example patent met "the requirement under 35 USC 112 first paragraph as providing an adequate written description of the claimed invention."

The patent's first claim, which is representative of the others, claimed all antibodies that bind to PCSK9 and block it from binding to LDL cholesterol:

> An isolated monoclonal antibody, wherein, when bound to PCSK9, the monoclonal antibody binds to at least one of the following residues . . . and wherein the monoclonal antibody blocks binding of PCSK9 to LDL[-]R.[167]

Amgen's first claim followed the same format outlined in the 1999 training materials: "an isolated antibody capable of binding to antigen X."[168] Moreover, Amgen taught "26 exemplary antibodies it identifies by their amino acid sequences," and thereby went beyond the 1999 training materials example specification, which only contemplates but does not describe example antibodies.[169]

But the Federal Circuit has slowly eroded the antibody exception since as late as *Centocor* in 2011.[170] In 2017, the Federal Circuit outright rejected the antibody exception the first time it considered *Amgen*: "the newly characterized antigen test flouts basic legal principles of the written description requirement."[171] The Federal Circuit's ruling foreshadowed the Supreme Court's treatment of judge-made exceptions to the patent statutes.

Instead of applying the newly characterized antigen test (i.e., following the 1999 training materials guidelines for antibody enablement), the Supreme Court held that "the specification must enable the full scope of the invention as defined by its claims."[172] Applying this rule, the Court reasoned that "Amgen has failed to enable all that it has claimed, even allowing for a reasonable degree of experimentation."[173] Here, Amgen closely followed the form of the antibody exception as outlined in the 1999 training materials, even going beyond the standard in some places. Yet the Supreme Court held that Amgen did not enable the total breadth of their claims. Although the Court's decision did not mention the antibody exception or the newly characterized antigen test, the antibody exception is no longer consistent with case law, and antibody patents will need to find a new way to claim a genus of antibodies.

167. U.S. Patent No. 8,829,165 (filed Apr. 10, 2013).

168. *See* Comm'r Pat., Revised Written Description Guidelines Training Materials 59 (1999), https://web.archive.org/web/20110928022825/http://www.uspto.gov/web/offices/pac/writtendescription.pdf.

169. *See id.*; *Amgen*, 598 U.S. at 612.

170. *Centocor*, 636 F.3d at 1352.

171. *Amgen*, 872 F.3d at 1378 (internal quotations omitted).

172. *Amgen*, 598 U.S. at 610 (citing 35 U.S.C. § 112).

173. *Id.* at 613.

2. *The Supreme Court's Decision in Amgen Discouraged Tech-Exceptionalism.*

Not only is the antibody exception no longer consistent with Supreme Court case law, but the Supreme Court took several opportunities in its *Amgen* decision to spurn future attempts at carving out similar exceptions to § 112.[174] For example, the Court noted that its maxim regarding patent enablement, "the more a party claims, the broader the monopoly it demands, the more it must enable," applies "whether the case involves telegraphs devised in the 19th century, glues invented in the 20th, or antibody treatments developed in the 21st."[175] The Court also emphatically agreed with Amgen's brief, which stated "the Patent Act provides a single, universal enablement standard for all invention[s]."[176]

Before the Supreme Court granted certiorari on *Amgen*, the Federal Circuit also discouraged tech-exceptionalism in its 2017 *Amgen* decision. As noted earlier, the court found that the newly characterized antigen test "flouts basic legal principles of the written description requirement."[177] But the court's disdain for tech-exceptionalism was not limited to the antibody exception: "we have generally eschewed judicial exceptions to the written description requirement based on the subject matter of the claims."[178] This is not to say that there is no way to create an altered standard to the enablement doctrine. According to the Federal Circuit opinion, an act of Congress could validly create "a special written description requirement for antibodies."[179] However, without an act of Congress to amend the written description requirement, antibody patents must conform to Section 112. As the Supreme Court said, antibodies patents "may involve a new technology, but the legal principle is the same."[180]

3. *The Supreme Court Endorsed the Full Scope Enablement Test*

Having unequivocally found that "[j]udges may no more subtract from the requirements for obtaining a patent that Congress has prescribed than they may add to them," the Court in *Amgen* returned to the text of § 112 for

174. Exceptionalism is the condition of being different from the norm. *Exceptionalism*, MERRIAM-WEBSTER DICTIONARY (2023).

175. *Amgen*, 598 U.S. at 613.

176. *Id.* at 614 (internal quotations omitted).

177. *Amgen*, 872 F.3d at 1378.

178. *Id.* at 1379.

179. *See id.* (interpreting the plant patent written description statute, 35 U.S.C. § 162, as "exempting plant patents from § 112 if the description is as complete as is reasonably possible") (internal quotations removed).

180. *Amgen*, 598 U.S. at 616.

guidance.[181] The Court held that "courts cannot detract from the basic statutory requirement that a patent's specification describe the invention in such full, clear, concise, and exact terms as to enable any person skilled in the art to make and use the invention."[182] Specifically, a "specification must enable the full scope of the invention as defined by its claims."[183] This test mirrors the Federal Circuit's full scope enablement test, which it has used to evaluate enablement across technology sectors. Federal Circuit cases such as *MagSil v. Hitachi Global Storage Technologies* in 2012 and *Idenix v. Gilead* in 2019 show that the Federal Circuit has been using the full scope enablement test for more than a decade before the *Amgen* decision.

In *MagSil*, the Federal Circuit found that an adequately enabled patent "must teach those skilled in the art how to make and use the full scope of the claimed invention without undue experimentation."[184] MagSil's patent claimed a sensor which, in part, can cause "a change in the resistance by at least 10%":

> 1. A device forming a junction having a resistance comprising:
>
> a first electrode having a first magnetization direction,
>
> a second electrode having a second magnetization direction, and
>
> an electrical insulator between the first and second electrodes, wherein applying a small magnitude of electromagnetic energy to the junction reverses at least one of the magnetization directions and causes a change in the resistance by at least 10% at room temperature.[185]

According to the way claim 1 was written, the court found that the patent "claim[ed] resistive changes from at least 10% up to infinity."[186]

The court, however, found the specification lacking. The court pointed to places in the specification where MagSil claimed an ideal sensor could "yield around a 24% change in resistance" and where MagSil taught, at the time of filing, "the best achievement was an 11.8% change."[187] The court concluded that the "patent specification only enables an ordinarily skilled artisan to achieve a small subset of the claimed range."[188] And the court continued by

181. *Id.* at 612.
182. *Id.* (citing 35 U.S.C. § 112) (internal quotations omitted).
183. *Id.* at 610 (citing 35 U.S.C. § 112).
184. MagSil Corp. v. Hitachi Glob. Storage Techs., Inc., 687 F.3d 1377, 1380 (Fed. Cir. 2012) (emphasis added).
185. *Id.* at 1379.
186. *Id.* at 1382.
187. *Id.*
188. *Id.* at 1384.

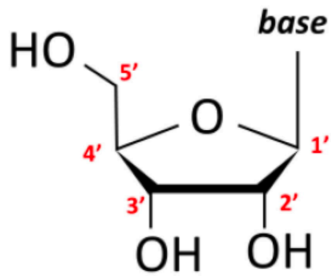
saying the specification in no way predicts, much less enables, "achievement of the modern values above 600% without undue experimentation, indeed without the nearly twelve years of experimentation necessary to actually reach those values."[189]

Similarly, in *Idenix*, the Federal Circuit found that Idenix's claim in a hepatitis C drug patent was invalid for lack of enablement because practicing the full scope of Idenix's claims would require "many, many thousands of candidate compounds, many of which would require synthesis and each of which would require screening" and found this level of experimentation to be undue.[190]

Idenix owned Patent No. 7,608,597, which claimed a small-molecule hepatitis C drug:

> A method for the treatment of a hepatitis C virus infection, comprising administering an effective amount of a purine or pyrimidine β-D-2'-methyl-ribofuranosyl nucleoside or a phosphate thereof, or a pharmaceutically acceptable salt or ester thereof.[191]

Idenix's claimed molecule "contain[ed] a sugar ring having five carbon atoms, numbered 1' (one prime) to 5' (five prime), as well as a base."[192] At each carbon, "substituent atoms or groups of atoms can be added in either the 'up' or 'down' position."[193] The structure illustrated in the court's opinion is reproduced below, "with a hydroxyl group (OH) shown attached at the 2'-down and 3'-down positions."[194]



Idenix claimed that the key to its drug was "having a methyl substitution ('CH3') at the 2' 'up' position of the molecule's sugar ring."[195] However, Gilead contended that this specification provided "no guidance in determining which

189. *Id.*
190. Idenix Pharms. LLC v. Gilead Scis. Inc., 941 F.3d 1149, 1163 (Fed. Cir. 2019).
191. *Id.* at 1155.
192. *Id.* at 1154.
193. *Id.*
194. *Id.*
195. *Id.*

of the billions of potential 2'-methyl-up nucleosides" effectively treats Hepatitis C.[196]

The Federal Circuit found that Idenix's patent did "not provide enough meaningful guidance or working examples, across the *full scope* of the claim, to allow a POSA to determine which 2'-methyl-up nucleosides would or would not be effective against HCV without extensive screening."[197]

Idenix and *MagSil* are just two cases from a broad jurisprudence that show the Federal Circuit applying the full scope enablement test to patents claiming technology other than antibodies. The Supreme Court's decision in *Amgen* endorsed the full scope enablement test for antibody patents, but the Federal Circuit has judged patents beyond antibody technologies under the same full scope enablement test used in *Amgen* for quite some time.

Under the current states of patent law and antibody technology, antibody researchers will likely be unable to claim an entire genus. Some legal scholars even argue that the genus claim is dead.[198] In *The Death of the Genus Claim*, Karshtedt, Lemley, and Seymore claim that it is "nearly impossible to maintain a valid genus claim" because courts "almost always hold them invalid."[199] This leaves antibody researchers only able to claim very narrowly around the antibodies they have physically synthesized and studied in a lab. However, some advancements in computer modeling technology may keep the genus claim alive.

B. UPCOMING TECHNOLOGICAL ADVANCES MAY STILL ALLOW BROAD GENUS CLAIMS UNDER *AMGEN*

Amgen identified twenty-six exemplary antibodies using their amino acid sequences in its patent specification, and the Supreme Court did "not doubt that Amgen's specification enable[d] the 26 exemplary antibodies."[200] But how far does the Court's logic go? Presumably, the Court would have no problem holding that Amgen's specification would have enabled twenty-seven, thirty, or even fifty exemplary antibodies if Amgen had identified all of them via their amino acid sequences. Sanofi argued that Amgen's patent, as written, claimed "potentially millions more undisclosed antibodies that perform these same functions."[201] What if Amgen had identified one million exemplary antibodies that bind to PCSK9? Currently the time and money it would take to identify

196. *Id.* at 1155.
197. *Id.* at 1162.
198. *See generally* Karshtedt et al., *supra* note 21.
199. *Id.* at 1.
200. *Amgen*, 598 U.S. at 612–13.
201. *Id.* at 603.

one million exemplary antibodies prohibit pharmaceutical developers from profitably undertaking this gargantuan exercise, but new technology could significantly lower the cost to identify large numbers of antibodies that perform a specific function.

Section IV.B.1 of this Note will discuss the technology needed to identify large numbers of antibodies that perform a specific function, and Section IV.B.2 will explore potential impacts on enablement of antibody genus claims through three hypothetical scenarios.

1. *Upcoming Developments in Computer Aided Protein Folding Modeling May Lower the Cost to Design and Develop Antibodies*

Google's DeepMind has released and continues to develop an artificial intelligence (AI)-based program called AlphaFold.[202] According to its website, "AlphaFold is an AI system . . . that predicts a protein's 3D structure from its amino acid sequence."[203] Before this development, antibody scientists had not been able to predict an antibody's 3D structure from an amino acid sequence with any reliability.[204] This new tool from Google and others like it have the power to make antibody drug development orders of magnitude faster and cheaper. Drug developers will have the ability to use models to predict whether a theoretical antibody will bind to a target antigen, where on the antigen the antibody will bind, and for how long. While drug developers will still need to run their prospective antibody treatments through clinical trials, AI protein folding models may lower the cost and time associated with developing prospective treatments.

2. *Developments in Computer Aided Protein Folding May Also Enable Broad Genus Claims*

Given the recent and predicted developments in AI technology, this Note identifies three hypotheticals in which antibody drug developers either use new technology to enable broad genus claims, like the ones in Amgen's patent and the 1999 Training Materials, or design around more narrow antibody claims. The first hypothetical proposes a drug developer using AI-assisted protein folding modeling to design-around a patent that claims relatively few antibodies. The other two hypotheticals look at how drug developers can use AI-based protein folding algorithms to patent broad genus claims. The

202. AlphaFold Protein Structure Database Home, ALPHAFOLD, https://alphafold.ebi.ac.uk/ (last visited Mar. 1, 2024).

203. *Id.*

204. *See supra* Section II.D.1.

hypotheticals rely on several assumptions about AI-based protein folding models and the state of antibody drug development:

- AI protein folding models can or will be able to adequately predict protein folding, protein three-dimensional structures, and protein function given a chain of amino acids.
- AI protein folding models will have a guarantee or percent certainty that the outputs of the program are accurate in the real world.
- A PHOSITA can now or will be able to synthesize an antibody from a chain of amino acids like the ones listed in Amgen's patent.
- AI protein folding models will be able to be imaged, i.e., researchers can save the state of an AI model on a particular day and time so that others can recreate experiments knowing the AI engine will not have changed or learned new information in the time between the two experiments.

a) Hypothetical 1: AI-Assisted Antibody Design Around

The first scenario implicating AI in enablement traces the path of an antibody drug company soon after *Amgen*. A company called notAmgen files a patent intending to cover a genus of antibodies. The company lists twenty-six. Under *Amgen*, it has a claim to those twenty-six antibodies, but not much else. A competitor company called notSanofi uses AI to substitute a few amino acids into the disclosed antibodies from the notAmgen patent and ensure that the new antibodies' three-dimensional structure still allows them to bind to the target antigen.[205]

Here, notSanofi has successfully designed around notAmgen's patent. Depending on the quality of the AI used to design around notAmgen's patent, notSanofi may have even discovered a more effective antibody treatment.

Under this fact pattern, antibody patent owners can only assert their patent rights over antibodies they have described in the patent specification. However, other antibody companies can input into AI tools the information the patent teaches about the antigen and the antibody genus and develop more antibody sequences to easily design around the patent. Although antibody companies can already design around patents, AI tools may make the practice easier, cheaper, and faster, and thus more widespread as the barrier to entry

205. Antibody researchers can do this without the use of AI technology, but the technology will make this process easier and cheaper.

lowers. If this scenario becomes the status quo, antibody developers will see patents as less-useful IP protection tools and may opt to keep antibodies and antigens as trade secrets.[206] In that case, the public would lose the knowledge transfer that the patent system facilitates.

b) Hypothetical 2: AI Protein Folding Model Assisted Enablement

In Hypothetical 2, a company, notAmgen, develops a new antibody drug. The company has successfully produced twenty-six monoclonal colonies that produce successfully binding antibodies. Next, it uses an AI program to extrapolate from those twenty-six working examples to find 10,000 more. notAmgen is confident it has found every amino acid sequence for every antibody that would theoretically bind to their identified antigen.

In this scenario, if notAmgen applies for a patent and claims the genus of antibodies that bind to antigen X with an affinity of at least Y at cite Z, it can now list all 10,000 theoretical examples that it found.[207] An enterprising patent prosecutor, however, decides it would be more prudent to enable all 10,000 antibodies by listing the AI engine, the inputs given to it, the training data the engine had access to, and the date on which the 10,000 theoretical antibodies were generated. Assuming the training data and the AI engine are readily available to the public, would the genus claim be adequately enabled?

This would certainly be a question for the USPTO and the courts. However, for the public to get their end of the bargain—in this case, teachings about new antibodies—(1) the public would need access to the AI engine in question, (2) the public would need access to the training data, and (3) the specification would need to teach how to use and interpret the data with the AI engine.

c) Hypothetical 3: AI Protein Folding Model Assisted Antibody Identification

More likely than disclosing the AI data as in Hypothetical 2, the patentee would not list the AI engine as a development tool so that the company can keep its data confidential instead of disclosing it. Patentees would consider maintaining confidentiality, especially if they developed their own AI tool, instead of using a publicly available one.

Thus, the third hypothetical is very similar to Hypothetical 2 except that, instead of listing the AI engine, the inputs given to it, the training data it had

206. The FDA might have more to say about keeping pharmaceuticals secret, but that is outside the scope of this Note.

207. There may be written description or utility issues with this method of disclosure, but this hypothetical is only focused on enablement.

access to, and the date on which the 10,000 theoretical antibodies were generated, the patent prosecutor lists all 10,000 amino acid sequences in the specification of the patent. Here, there are fewer questions of enablement.[208] All 10,000 species are well-described. However, there are some possible repercussions. For example, the USPTO would not appreciate the increasing size of patents. The patent agents only have so much time per patent and, listing 10,000 antibody sequences would not lighten their workload.

In this scenario, the patent drafter is no longer claiming a genus of antibodies. Instead, she is claiming 10,000 antibodies that all happen to have the same function. This scenario teaches the public about the claimed antibodies so theoretically the invention would be adequately described. Moreover, a PHOSITA would be enabled to make and use all 10,000 antibodies. And the patent owner would not need to worry about design-around. Everyone is happy, except for the patent examiner who gets assigned a 10,000-antibody patent. There are more ramifications if this hypothetical were replicated on a large scale. If all antibody patent examiners were slowed down by reading 10,000 amino acid sequences per patent, the patent system could experience massive delays. Perhaps AI can also help the USPTO read and examine patents.

V. CONCLUSION

Amgen was the final chapter in the Antibody Exception saga. From the first synthesized antibody in 1975 to *In re Wands* to *AbbVie*, the Federal Circuit has grappled with the complex doctrines of written description and enablement and tried to adjust them to fit a technology. However, *Amgen* marks the return to one disclosure standard for all technologies, in line with the text of the patent statute.

Moving forward, AI may again change the way the courts look at written description and enablement. The USPTO's proposed guidance on AI-enabled-patents should provide the first glimpse into how the doctrines will change in the future.[209]

208. The same questions regarding written description and utility from Hypothetical 2 crop up in Hypothetical 3.

209. *See* Comm'r Pat., Director's Blog: the Latest from USPTO Leadership, Latest Updates on Artificial Intelligence and Intellectual Property (Sept. 29, 2023).